Manuscripts considered — chondrosarcoma-mets-acetabular-fx-idh-unknown-y34r

Master inventory: every paper reviewed in this case

Generated 2026-08-24

Manuscripts considered — chondrosarcoma-mets-acetabular-fx-i

Master inventory: 30 clinical (27 included, 3 considered & excluded) + 30 pre-clinical (28 included, 2 considered & excluded) + 21 additional trial publications/registrations. One entry per manuscript, sorted by year (newest first). Excluded entries are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

Remiszewski/Czarnecka (2026) — Crit Rev Oncol Hematol

Reference:PMID 41895355 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Chemotherapy in dedifferentiated chondrosarcoma (review)
- **Indication / model:** Dedifferentiated chondrosarcoma, neoadjuvant to palliative (any); Narrative synthesis of the dedifferentiated-specific literature; metastases in ~40-80%, contemporary 5-year OS ~7-24%.
- **Design:** Narrative review
- **Effect size:** ~20 % — First-line ORR in advanced disease (synthesized)
- **Toxicities:** Review notes lower pathological response rates than osteosarcoma and substantial haematological toxicity with osteosarcoma-type regimens; no primary AE table.
- **Case fit:** strong
- **Notes:** Most current dedifferentiated-specific synthesis; source of the ORR ~20% / PFS 4-5 months figures carried from intake. Its olutasidenib line (DDCS mPFS 1.5 months) is the only published IDH-inhibitor efficacy figure in dedifferentiated disease and argues the IDH axis is weaker in her histology than in conventional CS.

Kloker/Deinzer (2026) — BMC Cancer

Reference:PMID 42251292 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** B7-H3 membranous expression across sarcoma subtypes (target-density measurement)
- **Indication / model:** two-centre cohort of 133 pre-treatment human bone and soft-tissue sarcoma samples, centrally stained B7-H3 IHC with H-score
- **Design:** H-score separates positivity from density. Both the CAR and the ADC literature make efficacy a function of surface antigen per cell, so a 'percent positive' figure and an H-score are not interchangeable.
- **n:** n=133 (103 soft-tissue, 30 bone sarcoma)
- **Effect size:** moderate — B7-H3 was expressed in 97% of sarcomas, but chondrosarcoma had the lowest median H-score of any subtype (60) against 175 for osteosarcoma and 202.5 for pleomorphic liposarcoma. H-score rose with histological grade across subtypes and predicted event-free but not overall survival.
- **Variance:** vs cross-subtype comparison; four B7-H3-negative samples; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Human tissue survey, not a model system, and the chondrosarcoma subgroup within the 30 bone sarcomas is small with no dedifferentiated breakdown reported. The grade correlation leaves open that a dedifferentiated component scores higher than the subtype median.

Ao/Pollack (2026) — Cancer Immunol Res

Reference:PMID 41719158 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Membrane-bound IL-15-engineered TILs from chondrosarcoma (cytoTIL15)
- **Indication / model:** TILs expanded from immune-excluded human chondrosarcoma biopsies; autologous 3D tumour spheroid killing assays; spatial profiling of the TME
- **Design:** Collagenous matrix and myeloid infiltration exclude lymphocytes to the tumour periphery; membrane-bound IL-15 lowers the TCR signalling threshold of TIL clonotypes so the same T cells infiltrate and kill.
- **n:** not stated per patient in the abstract
- **Effect size:** moderate — Spatial profiling put ubiquitous collagen and myeloid infiltration forward as the resistance mechanisms, with lymphocytes largely restricted to peripheral regions — which is also where a TIL harvest would have to sample. Engineered TILs expanded from paucicellular chondrosarcoma biopsies and killed autologous 3D tumour models without exogenous IL-2.
- **Variance:** vs conventional IL-2-dependent TIL expansion; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Ex-vivo and 3D-model endpoints only, no in-vivo efficacy arm, and the work is sponsored by the company developing the platform.

Johnson ML/Doi T (2026) — Lancet Oncol

Reference:PMID 41926962 · **Type:** trial · **Inclusion:** included

- **Intervention:** Ifinatamab deruxtecan (B7-H3-directed antibody-drug conjugate)
- **Indication / model:** Advanced treatment-refractory solid tumours including sarcoma (IDeate-PanTumor01 dose escalation) (2L+ (treatment-refractory)); none required; B7-H3 expression not an entry criterion
- **Design:** First-in-human two-part phase 1/2 dose escalation
- **n:** 97
- **Effect size:** — — Safety; antitumour activity by RECIST 1.1 at >=4.8 mg/kg
- **Case fit:** weak
- **Notes:** (toxicity table not extracted; primary endpoint not extracted)

Tap/Trent (2025) — Clin Cancer Res

Reference:PMID 40100120 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Ivosidenib, long-term follow-up (conventional chondrosarcoma subset)
- **Indication / model:** Advanced mutant-IDH1 conventional chondrosarcoma (any); 21 chondrosarcoma patients dosed in the phase 1; efficacy reported only for the 13 with conventional histology. Database lock March 2024.
- **Design:** Long-term follow-up of phase 1 (NCT02073994)
- **n:** 13
- **Effect size:** 23.1 % — ORR, conventional subset
- **Toxicities:** any treatment-related AE (15/21, 71.4%); any serious AE (6/21, 28.6%)
- **Case fit:** partial
- **Notes:** Source of the ORR 23.1% figure carried from intake - it belongs to the 13-patient conventional subset, not to all 21 dosed. The 8 non-conventional patients have no reported subtype-level efficacy in either paper; do not read this ORR as dedifferentiated evidence. The successor phase 3 (NCT06127407) enrols conventional histology only.

Cross/Van Loo (2025) — Genome Biol

Reference:PMID 41299743 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Mutant IDH1/2 loss during metastatic evolution
- **Indication / model:** Metastatic central chondrosarcoma (genomic evolution, PEACE autopsy programme) (any); Multi-region primary and metastatic sampling of central chondrosarcoma within the PEACE consortium.
- **Design:** Genomic evolution study (multi-region/autopsy sequencing)
- **Effect size:** mutant IDH1 recurrently lost in metastatic disease — Recurrent loss of mutant IDH1 in metastases
- **Toxicities:** Not applicable (no therapeutic exposure).
- **Case fit:** partial
- **Notes:** Bears directly on tissue choice for her IDH testing: a metastasis may have lost the mutation the primary carries, and an IDH-directed drug may be targeting a clone that no longer needs it. Test the primary or, ideally, both compartments.

Palmerini/Martin-Broto (2025) — Cancer

Reference:PMID 39540661 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Sunitinib plus nivolumab (IMMUNOSARC, bone-sarcoma cohort)
- **Indication / model:** Advanced progressing bone sarcoma (17 osteosarcoma, 14 chondrosarcoma of which 4 dedifferentiated, 8 Ewing, 1 UPS) (2L+); 40 treated, 38 evaluable by central review; documented progression required; median age 47.
- **Design:** Multicentre single-arm phase 2 (IMMUNOSARC, NCT03277924)
- **n:** 40
- **Effect size:** 42 % — 6-month PFS rate (central review)
- **Variance:** 95% CI 27–58
- **Toxicities:** neutropenia G>=3 (6/40, 15%); hypertension G>=3 (6/40, 15%); anaemia G>=3 (5/40, 12.5%); ALT/AST elevation G>=3 (5/40, 12.5%); pneumonitis G>=3 (1/40, 2.5%); discontinuation for toxicity (7/40, 17%)
- **Case fit:** partial
- **Notes:** The published readout behind trial-screen row NCT03277924: one of the four dedifferentiated chondrosarcoma patients had a PR. Endpoint met, but a treatment-related grade 5 pneumonitis and 17% toxicity discontinuation matter for an assumed-ECOG-2 patient.

Masunaga/Kawai (2025) — Int Orthop

Reference:PMID 40613902 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Resection of the primary in metastatic chondrosarcoma (JNBSTR)
- **Indication / model:** Chondrosarcoma metastatic at initial diagnosis (grade 2-3 conventional, dedifferentiated, mesenchymal) (1L); 62 patients, 2001-2022, Japanese National Bone and Soft Tissue Tumour Registry; propensity matching on number of metastatic organs, metastasectomy and chemotherapy.
- **Design:** Retrospective registry cohort with propensity score matching
- **n:** 62
- **Effect size:** 51.0% vs 19.3% % — 2-year disease-specific survival, resected vs not
- **Variance:** p=0.005
- **Toxicities:** No surgical-morbidity reporting.
- **Case fit:** partial

- **Notes:** The pro-resection signal on the oncologic question, but the 2-month median in the unresected arm signals residual confounding by fitness and disease pace that propensity matching cannot remove. Pools conventional, dedifferentiated and mesenchymal histologies without a dedifferentiated subgroup estimate.

Masunaga/Kawai (2025) — SICOT J

Reference:PMID 40079610 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Prognostic-factor analysis, DDCS (JNBSTR)
- **Indication / model:** Dedifferentiated chondrosarcoma (national registry) (any); 132 pathologically confirmed DDCS, 2006-2022; 34 metastatic at diagnosis, 98 localized.
- **Design:** Retrospective national-registry cohort
- **n:** 132
- **Effect size:** 9.7% vs 37.1% % — 5-year disease-specific survival, metastatic vs localized
- **Variance:** p=<0.0001
- **Toxicities:** No AE reporting (registry).
- **Case fit:** strong
- **Notes:** Contemporary dedifferentiated-specific prognosis: metastatic-at-diagnosis 5-year DSS under 10%. Its favourable adjuvant-chemo HR is the outlier among the chemo comparisons and is localized-disease, registry-level evidence; treat the chemo signal as unresolved rather than settled either way.

Dong/Bai (2025) — Crit Rev Oncol Hematol

Reference:PMID 40889733 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Proton beam therapy for chondrosarcoma (meta-analysis)
- **Indication / model:** Chondrosarcoma treated with proton therapy (six studies, predominantly skull base) (1L); 282 patients across 6 studies; median cohort size 47; mean of median ages 38.9; largely definitive/postoperative treatment of unresected or incompletely resected non-metastatic disease.
- **Design:** Systematic review and meta-analysis of observational cohorts
- **n:** 282
- **Effect size:** 95 % — 5-year local control (pooled)
- **Variance:** 95% CI 91–99
- **Toxicities:** proton-therapy-related toxicity
- **Case fit:** weak
- **Notes:** Why the particle class exists for this radioresistant histology: local control above 90% at 5 years where photons historically fail. Nearly all data are skull-base, low-to-intermediate-grade, non-metastatic cohorts, so the figures describe the modality's ceiling, not her pelvic, high-grade, metastatic situation. Pooling-model CIs exceeding 100% flag the analysis's limits.

Barisa/Anderson (2025) — Nat Commun

Reference:PMID 40764385 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Anti-B7-H3 CAR constructs, avidity versus antigen-density threshold
- **Indication / model:** three clinical-candidate B7-H3 binders across tumoroid re-stimulation assays, BEHAV3D single-cell video microscopy, and in-vivo models
- **Design:** Cellular avidity, not monomeric affinity, sets the antigen-density floor at which a CAR triggers;

higher-avidity constructs hold longer CD8 contacts with dim targets and keep expanding.

- **n:** 3 binder constructs; per-arm animal counts not stated in the abstract
- **Effect size:** moderate — Functional avidity predicted the B7-H3 density threshold for effector function, with the better constructs giving longer cumulative CD8 CAR-T/tumour contact times, more CAR-T expansion and sustained control. Lower checkpoint-receptor expression did not correlate with better anti-tumour function.
- **Variance:** vs comparison across the three binders of differing affinity; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Paediatric neuroblastoma and related models, no sarcoma-of-bone arm. Relevant because it defines what would have to be engineered around for an antigen-dim tumour, not because it tested one.

Quoniou/Peyrode (2025) — Med Oncol

Reference:PMID 40960554 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Mifamurtide (TLR4 agonist) with doxorubicin
- **Indication / model:** CH2879/THP-1 3D co-culture spheroids and CH2879 murine xenograft
- **Design:** TAMs can reach half the tumour mass in chondrosarcoma and sit predominantly in an M2-like state; a TLR4 agonist repolarises them, and the chemosensitivity of the tumour compartment changes as a consequence.
- **n:** not stated per arm in the abstract
- **Effect size:** moderate — Mifamurtide increased the chemosensitivity of CH2879/THP-1 spheroids, and in the CH2879 xenograft the combination with doxorubicin improved antitumour efficacy while reducing intermediate-stage TAMs.
- **Variance:** vs doxorubicin alone; mifamurtide alone; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Conventional grade III line, no dedifferentiated model, and mifamurtide's approved indication is paediatric non-metastatic osteosarcoma, so the regulatory route in this setting is not established.

—/— (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Envafolimab plus doxorubicin and ifosfamide
- **Indication / model:** Advanced soft tissue sarcoma, first line (1L (no previous systemic treatment)); none required
- **Design:** Single-arm phase 2
- **n:** 25
- **Effect size:** — — Progression-free survival
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** HS-20093 with anlotinib, epirubicin or adebrelimab
- **Indication / model:** Locally advanced or metastatic osteosarcoma and soft tissue sarcoma (1L in the cohort 1 dose-expansion; 2L+ elsewhere); none required
- **Design:** Open-label phase 1b combination study (ARTEMIS-103)
- **n:** 448
- **Effect size:** — — Maximum tolerated dose for the combinations
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Tebentafusp (gp100 peptide-HLA-A*02:01 directed ImmTAC)
- **Indication / model:** Advanced clear cell sarcoma, HLA-A*02:01 positive (2L+ per protocol); HLA-A*02:01 by genotyping
- **Design:** Randomised phase 2 against physician’s choice
- **n:** 47
- **Effect size:** — — Disease response
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Tsukamoto/Errani (2024) — Curr Oncol

Reference:PMID 38275833 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Adjuvant chemotherapy (systematic review, localized DDCS)
- **Indication / model:** Localized dedifferentiated chondrosarcoma after surgery (adj); 556 patients with localized DDCS pooled from 11 retrospective non-randomized studies.
- **Design:** Systematic review with pooled odds ratio (PubMed/Embase/CENTRAL)
- **n:** 556
- **Effect size:** 1.25 OR — 5-year survival odds ratio, adjuvant chemo vs surgery alone
- **Variance:** 95% CI 0.8–1.94; p=0.324
- **Toxicities:** Review of survival outcomes only; no pooled AE data.
- **Case fit:** partial
- **Notes:** Directly answers the intensification question for localized disease: no pooled survival benefit from adjuvant chemotherapy, with a possible peripheral-subtype exception. All 11 inputs are retrospective; nothing here is randomized.

—/— (2024) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Ivosidenib 500 mg orally once daily
- **Indication / model:** Locally advanced or metastatic conventional chondrosarcoma, IDH1-mutant (1L-2L); IDH1 R132C/L/G/H/S by central testing on primary or metastatic tissue

- **Design:** Randomised, double-blind, placebo-controlled phase 3
- **n:** 136
- **Effect size:** — — PFS by blinded independent central review in grade 1 and 2 participants
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Enasidenib
- **Indication / model:** IDH2-mutated malignant sinonasal and skull base tumours, including chondrosarcoma (2L+ (prior systemic therapy in the recurrent/metastatic setting required)); Somatic IDH2 R140 or R172
- **Design:** Single-arm phase 2, multi-histology
- **n:** 40
- **Effect size:** — — Progression-free survival
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Doxorubicin plus pembrolizumab
- **Indication / model:** Metastatic or unresectable dedifferentiated liposarcoma, undifferentiated pleomorphic sarcoma and related poorly differentiated sarcomas (1L (no prior systemic therapy for advanced disease)); none required
- **Design:** Randomised phase 3
- **n:** 365
- **Effect size:** — — Progression-free survival
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** MGC026 (B7-H3-directed antibody-drug conjugate)
- **Indication / model:** Advanced solid tumours including sarcoma (2L+ in practice (post-standard-therapy population)); none required; archival or fresh tissue mandatory
- **Design:** First-in-human phase 1/1b dose escalation and cohort expansion
- **n:** 250
- **Effect size:** — — Incidence of adverse events
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not

extracted)

—/— (2024) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Risvutatug rezetecan (GSK5764227), B7-H3-directed antibody-drug conjugate, alone or with platinum or checkpoint blockade
- **Indication / model:** Advanced solid tumours (2L+ (progressed on or intolerant to available standard therapy)); none required
- **Design:** Phase 1 monotherapy and combination dose escalation and expansion (EMBOLD)
- **n:** 845
- **Effect size:** — — Adverse events and dose-limiting toxicities
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Bui/Davis (2023) — Cancers (Basel)

Reference:PMID 37174084 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Systemic chemotherapy (five-centre DDCS cohort)
- **Indication / model:** Dedifferentiated chondrosarcoma, localized and metastatic (any); 74 DDCS patients from five academic sarcoma centres, 2004-2022; 58% received systemic therapy, predominantly in the metastatic setting.
- **Design:** Retrospective multi-institutional cohort
- **n:** 74
- **Effect size:** 9 % — ORR to systemic therapy
- **Toxicities:** No per-term AE reporting (retrospective chart review).
- **Case fit:** strong
- **Notes:** Modern US/UK cohort with the least favourable chemo figures in the dossier: ORR 9% despite osteosarcoma-protocol-style 1L regimens, and metastatic median OS 7.2 months with 0% alive at 5 years. Read against Italiano 20.5%, it brackets the honest range for her 1L expectation. AE data genuinely absent from source.

Dermawan/Hameed (2023) — Cancer Res Commun

Reference:PMID 36926116 · **Type:** clinical · **Inclusion:** included

- **Intervention:** IDH1/2 mutation prevalence and biology in dedifferentiated chondrosarcoma
- **Indication / model:** Dedifferentiated vs conventional chondrosarcoma (molecular profiling) (any); 18 DDCS (with macrodissected well-differentiated components) vs 55 conventional CS by targeted sequencing; 34 DDCS vs 94 conventional in methylation/expression analyses.
- **Design:** Molecular cohort study (MSK targeted sequencing + methylation/expression)
- **n:** 73
- **Effect size:** 71% DDCS vs 36% conventional % — IDH1/2 mutation prevalence
- **Toxicities:** Not applicable (no therapeutic exposure).
- **Case fit:** strong

- **Notes:** Justifies IDH1/2 testing in her workup: prevalence is higher in dedifferentiated than conventional disease. The same paper cautions that the IDH-associated methylation phenotype is partially lost at dedifferentiation and that IDH1-inhibitor trials showed worse outcomes in DDCS, so a positive test opens an option without promising conventional-CS-level benefit.

Bui/Davis (2023) — Cancers (Basel)

Reference:PMID 37174084 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Immune checkpoint inhibitors in DDCS (five-centre cohort, ICI subset)
- **Indication / model:** Advanced dedifferentiated chondrosarcoma (2L+); ICI-treated subset of the 74-patient DDCS cohort; the pembrolizumab responder was PD-L1 TPS 7% / CPS 8, microsatellite-stable.
- **Design:** Retrospective multi-institutional cohort (ICI-treated subset)
- **n:** 11
- **Effect size:** 1 PR among 5 pembrolizumab-treated — Best response to ICI
- **Toxicities:** No per-term AE reporting (retrospective).
- **Case fit:** strong
- **Notes:** Second row from the Bui cohort, split out because the ICI subset is a different intervention from the chemotherapy rows. A PD-L1-low, MSS responder cautions against gating checkpoint consideration on high PD-L1 alone.

Pathmanapan/Alman (2023) — Cell Rep

Reference:PMID 37267108 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Mutant IDH biology: glycogen dependency (founder-event context)
- **Indication / model:** mutant Idh1 mice developing enchondromas, murine growth plate, and human mutant-IDH chondrosarcoma cells in vitro and in vivo
- **Design:** Mutant IDH1 partners with HIF1alpha to divert carbon into glycogen, a deposit found only in mutant cells, so the metabolic consequence of the mutation persists as a dependency separate from D-2-HG-driven methylation.
- **n:** not stated per arm in the abstract
- **Effect size:** moderate — Glycogen deposition appeared exclusively in mutant cells from IDH-mutant chondrosarcomas and Idh1-mutant murine growth plates. Blocking glycogen utilisation changed tumour cell behaviour, downstream energetics and tumour burden both in vitro and in vivo.
- **Variance:** vs IDH wild-type cells and wild-type murine growth plate; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** The mouse model produces enchondroma, the benign precursor, not dedifferentiated chondrosarcoma; no clinical-grade glycogen-pathway agent exists.

—/— (2023) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** HS-20093 (B7-H3-directed antibody-drug conjugate)
- **Indication / model:** Relapsed or refractory osteosarcoma and other sarcomas (2L+ (progression on first-line systemic treatment required)); none required; fresh or archival tissue mandatory

- **Design:** Multicentre open-label phase 2 (ARTEMIS-002)
- **n:** 170
- **Effect size:** — — Objective response rate
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Cranmer/Wagner (2022) — Clin Orthop Relat Res

Reference:PMID 34648466 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Perioperative chemotherapy (SEER, dedifferentiated chondrosarcoma)
- **Indication / model:** Localized dedifferentiated chondrosarcoma treated with cancer-directed surgery (adj); 185 SEER patients 2000-2016 after excluding distant disease at diagnosis, non-bone primaries and unresected patients.
- **Design:** Retrospective population-based cohort (SEER), multivariable Cox with missing-data adjustment
- **n:** 185
- **Effect size:** 0.75 HR — OS hazard ratio, chemotherapy vs none
- **Variance:** 95% CI 0.49–1.12; p=0.16
- **Toxicities:** SEER carries no AE data.
- **Case fit:** partial
- **Notes:** The largest chemo-vs-none comparison in this histology finds no OS benefit, though the CI is wide enough to hide a real effect and SEER chemotherapy coding is known to misclassify. Localized disease only, so it bears on the intensification question rather than on her metastatic 1L decision. No AE data possible from SEER.

Singh/Carr-Ascher (2022) — Front Oncol

Reference:PMID 36465334 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pembrolizumab (case report, sustained CR)
- **Indication / model:** Metastatic PD-L1-positive dedifferentiated chondrosarcoma (any); Single patient with metastatic PD-L1-positive DDCS.
- **Design:** Case report
- **n:** 1
- **Effect size:** CR sustained 24 months — Complete response, sustained
- **Toxicities:** No structured AE reporting (case report).
- **Case fit:** strong
- **Notes:** An n-of-1 best case, not an effect estimate; kept because it is one of very few documented durable ICI remissions in exactly her disease and it anchors the plausibility ceiling for the checkpoint axis.

Gonzalez/Pretell-Mazzini (2022) — J Bone Oncol

Reference:PMID 36246299 · **Type:** clinical · **Inclusion:** excluded — Overlapping SEER cohort: appendicular-only dedifferentiated analysis drawn from the same SEER base as Cranmer 2022 and Song 2019, which already carry the chemotherapy and resection questions; excludes the axial/pelvic anatomy that is hers.

- **Intervention:** Appendicular dedifferentiated chondrosarcoma (SEER)

- **Notes:** Excluded: Overlapping SEER cohort: appendicular-only dedifferentiated analysis drawn from the same SEER base as Cranmer 2022 and Song 2019, which already carry the chemotherapy and resection questions; excludes the axial/pelvic anatomy that is hers.

Yagi/Ogawa (2022) — Cancers (Basel)

Reference:PMID 35454915 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Proton versus carbon-ion RBE across sarcoma cell lines
- **Indication / model:** four human sarcoma lines (MG63 osteosarcoma, HT1080 fibrosarcoma, SW872 liposarcoma, SW1353 chondrosarcoma) and three normal-tissue lines
- **Design:** RBE is the number that converts a physical dose into a biological one; if it varies by cell type, the fixed 1.1 and 2.1 used in planning systems are averages rather than facts about a given tumour.
- **n:** 7 cell lines under one standardised colony-formation protocol
- **Effect size:** strong — Mean RBE was 1.08 +/- 0.11 for protons and 2.08 +/- 0.36 for carbon ions, close to the 1.1 and 2.13 used clinically. Between-line spread reached 34% for protons and 32% for carbon ions among the sarcoma lines, and up to 48% between two normal lines under carbon ions.
- **Variance:** vs gamma-ray irradiation as the reference radiation; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** SW1353 is the only chondrosarcoma line and it is conventional grade II; the physical rationale for carbon over proton (roughly a doubling of biological effect) is quantified here, the histology-specific part is not.

Yamato/Agatsuma (2022) — Mol Cancer Ther

Reference:PMID 35149548 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** DS-7300a (ifinatamab deruxtecan), B7-H3-directed DXd ADC
- **Indication / model:** human cancer cell lines in vitro; cell-line xenograft and patient-derived xenograft mouse models; PK in cynomolgus monkey; toxicology in rat and monkey
- **Design:** Antibody binding to B7-H3 internalises a cleavable DXd topoisomerase-I payload, which raises phospho-CHK1 and cleaved PARP in the target cell.
- **n:** multiple xenograft and PDX models; per-arm counts not stated in the abstract
- **Effect size:** strong — DS-7300a inhibited growth of B7-H3-expressing lines but not B7-H3-negative ones, and produced antitumour activity in high-B7-H3 xenografts and PDX models of several tumour types. It was stable in circulation and tolerated in rats and monkeys.
- **Variance:** vs B7-H3-negative cancer cell lines and vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Every efficacy model is explicitly high-B7-H3; there is no chondrosarcoma model and no low-expression efficacy arm, so this does not address the expression tier a chondrosarcoma would most likely sit at.

—/— (2022) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** NY-ESO-1-directed TCR-engineered autologous T cells
- **Indication / model:** Advanced soft tissue sarcoma, NY-ESO-1 positive (2L+ in practice); NY-ESO-1 expression;

HLA-A*02:01 restriction implicit in the TCR construct
- **Design:** Single-arm investigator-initiated cell-therapy study
- **n:** 20
- **Effect size:** — — Adverse events and serious adverse events
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Hompland/Smeland (2021) — Eur J Cancer

Reference:PMID 33990016 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Osteosarcoma-type intensive chemotherapy (EURO-B.O.S.S.)
- **Indication / model:** Dedifferentiated chondrosarcoma, localized (60%) and metastatic (40%), age 42-65 (1L); 57 DDCS patients fit for intensive chemotherapy in a prospective non-randomized protocol; surgical complete remission achieved in 36 (63%).
- **Design:** Prospective non-randomized multicentre study (EURO-B.O.S.S. DDCS cohort)
- **n:** 57
- **Effect size:** 24 months — Median OS
- **Variance:** 95% CI 22–25
- **Toxicities:** treatment-related death G5 (0/57, 0%); any toxicity leading to <6 of 9 cycles delivered (18/57, 30%)
- **Case fit:** partial
- **Notes:** The best prospective evidence for osteosarcoma-type intensification in her histology and age band, and its metastatic 5-year OS of 29% is the most favourable figure in the metastatic DDCS literature - but the cohort is selected for chemo-fitness and the study has no comparator, so it cannot show intensification beats standard STS regimens. Per-term AE table is in the paywalled full text; abstract carries the headline rates, no PMC version, no CT.gov results record.

Nota/Schwab (2021) — Front Oncol

Reference:PMID 33912442 · **Type:** clinical · **Inclusion:** included

- **Intervention:** TIL, HLA, PD-1/PD-L1 and B7-H3 expression (biomarker context)
- **Indication / model:** Conventional and dedifferentiated chondrosarcoma (tissue microarray) (any); 52 conventional and 24 dedifferentiated specimens (MGH).
- **Design:** Immunohistochemical biomarker study
- **n:** 76
- **Effect size:** CD8+ TILs 90%; B7-H3 96%; PD-1 53%; PD-L1 33% % — Immune-marker prevalence, dedifferentiated subtype
- **Toxicities:** Not applicable (no therapeutic exposure).
- **Case fit:** strong
- **Notes:** Immunogenic-phenotype evidence for the dedifferentiated variant, and the only expression data behind the several B7-H3-directed trials the screener kept - none of which has published chondrosarcoma clinical results. One author was a Merck employee.

Sambri/Bianchi (2021) — J Orthop Sci

Reference:PMID 32564907 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Surgery in DDCS of the limbs, pathological-fracture analysis
- **Indication / model:** Dedifferentiated chondrosarcoma of the limbs (1L); 175 adults (median 66); 49 (28.0%) metastatic at diagnosis; 39 (22.3%) with pathological fracture; 69% femoral.
- **Design:** Retrospective single-institution cohort (Rizzoli)
- **n:** 175
- **Effect size:** no difference — Disease-specific survival, fracture vs no fracture (localized)
- **Variance:** p=0.638
- **Toxicities:** No per-term AE/complication table.
- **Case fit:** partial
- **Notes:** The direct dedifferentiated-specific answer on pathological fracture: in limb disease it did not worsen survival or local control, against Grimer 2007 where fracture was a poor prognostic factor. Limb data - a pelvic fracture is a different structural problem, and her metastatic status dominates prognosis regardless (10-year DSS 8.6%).

Li/Ye (2021) — Clin Cancer Res

Reference:PMID 34426437 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Immune classification of conventional chondrosarcoma (CyTOF multi-omics)
- **Indication / model:** fresh human conventional chondrosarcoma tumours profiled by CyTOF, whole-exome sequencing and flow cytometry, with mechanistic follow-up
- **Design:** IDH mutation raised chemokine levels and immune-cell infiltration in otherwise immune-inactivated tumours, which offers a route by which genotype rather than PD-L1 shapes who is infiltrated.
- **n:** 98 fresh tumours collected; 22 fully profiled; 12 patients received PD-1 antibody
- **Effect size:** moderate — Three immune phenotypes emerged: G-MDSC-dominant, immune-exhausted, and immune-desert. The immune-rich subtypes were characterised by IDH mutation and high grade, and all 3 of 12 PD-1-treated patients with disease control were retrospectively classified as immune-exhausted.
- **Variance:** vs cross-subtype comparison among the three immune phenotypes; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Conventional chondrosarcoma only, so it does not directly cover the dedifferentiated component; the 3-of-12 clinical correlation is retrospective and hypothesis-generating.

—/— (2021) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** IACS-6274 (glutaminase-1 inhibitor), alone or with bevacizumab and paclitaxel
- **Indication / model:** Advanced solid tumours; chondrosarcoma is a named dose-escalation population (not specified for the chondrosarcoma population); none required for the chondrosarcoma cohort
- **Design:** Phase 1 dose escalation and expansion, monotherapy and combinations
- **n:** 54
- **Effect size:** — — Incidence of adverse events
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2021) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Carbon-ion radiotherapy, proton therapy or surgery
- **Indication / model:** Newly diagnosed pelvic sarcoma involving bone, chondrosarcoma named (primary local therapy); none required
- **Design:** Prospective comparative-effectiveness trial of carbon-ion therapy, surgery and proton therapy
- **n:** 72
- **Effect size:** — — Change in functional outcome; proportion with local recurrence
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Tap/Trent (2020) — J Clin Oncol

Reference:PMID 32208957 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Ivosidenib (mutant-IDH1 inhibitor), phase 1 chondrosarcoma cohort
- **Indication / model:** Advanced mutant-IDH1 chondrosarcoma (any); 21 patients (12 escalation, 9 expansion); 11 had prior systemic therapy; histology includes conventional and non-conventional chondrosarcoma. IDH1 mutation required.
- **Design:** Phase 1 multicentre open-label dose-escalation/expansion (NCT02073994, AG120-C-002)
- **n:** 21
- **Effect size:** 5.6 months — Median PFS
- **Variance:** 95% CI 1.9–7.4
- **Toxicities:** any TEAE G>=3 (12/21, 57.1%); hypophosphatemia (treatment-related) G>=3 (1/21, 4.8%); fatigue (9/21, 42.9%); diarrhea (7/21, 33.3%); nausea (6/21, 28.6%); QT prolongation G1-2 (5/21, 23.8%)
- **Case fit:** partial
- **Notes:** The 21 dosed patients were not restricted to conventional histology, so ivosidenib has been given to non-conventional chondrosarcoma - the drug is not untried there, its subtype-level response rate is simply unreported. Requires a documented IDH1 mutation, which she does not yet have. AE table extracted from PMC full text (PMC7238491).

Takenaka/Imai (2020) — Cancer

Reference:PMID 32658315 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Carbon-ion radiotherapy, unresectable pelvic bone sarcoma - late complications and function
- **Indication / model:** Unresectable pelvic bone sarcoma (long-term survivors) (1L); 29 patients without local recurrence or metastasis, drawn from 112 CIRT-treated; mean follow-up 93 months; periacetabular tumours analysed as a risk factor.
- **Design:** Retrospective complication and functional-outcome cohort
- **n:** 29
- **Effect size:** femoral head necrosis 37%; pelvic fracture 48%; neurological deficit 52% % — Late complication rates
- **Toxicities:** femoral head necrosis (n=29, 37%); pelvic insufficiency fracture (n=29, 48%); neurological deficit (n=29, 52%)

- **Case fit:** partial
- **Notes:** The toxicity ledger for pelvic CIRT, in survivors only - a best-case denominator. Her lesion is periacetabular, the exact subgroup with the highest femoral-head-necrosis risk and worst function scores; and she already has a fractured, irradiation-candidate pelvis. Mixed pelvic bone sarcomas, not chondrosarcoma-specific.

Kim/Kim (2020) — Nat Commun

Reference:PMID 33024108 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** HIF-2alpha inhibition plus chemotherapy
- **Indication / model:** chondrosarcoma tumour xenograft mouse models plus weighted correlation network analysis of human chondrosarcoma biopsies
- **Design:** HIF-2alpha (EPAS1) sits upstream of the gene module that separates high-grade from low-grade chondrosarcoma, so blocking it removes the transcriptional programme for growth, invasion and metastasis rather than a single downstream effector.
- **n:** not stated per arm in the abstract
- **Effect size:** strong — HIF-2alpha was elevated in high-grade biopsies and EPAS1 amplification tracked with poor prognosis; knocking it down cut tumour growth, local invasion and metastasis in xenografts. Pharmacological HIF-2alpha inhibition combined with chemotherapy was synergistic for apoptosis and removed the malignancy signature in mice.
- **Variance:** vs vehicle and single-agent chemotherapy; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** No dedifferentiated-specific arm, and the inhibitor and dosing are not identified in the abstract, which limits how far the dose relevance can be judged.

Girard/Boumédiene (2020) — J Bone Oncol

Reference:PMID 32211283 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Carbon-ion versus X-ray irradiation of chondrosarcoma lines
- **Indication / model:** five human chondrosarcoma cell lines (CH2879, SW1353, JJ012, FS090, 105KC) in normoxia and chronic hypoxia
- **Design:** High-LET carbon ions produce clustered, hard-to-repair DNA lesions, so cell kill is less dependent on the repair capacity and cell-cycle position that make photons ineffective here.
- **n:** 5 cell lines; survival curves, gammaH2AX, Apo2.7, SA-beta-gal, whole-exome sequencing
- **Effect size:** moderate — Carbon ions caused more DNA damage and more cell death than X-rays across the panel, though the magnitude varied by line. X-ray response split cleanly — apoptosis and senescence in CH2879, one or the other in SW1353 and JJ012, no death at all in FS090 and 105KC — with p21 tracking senescence. Chronic hypoxia did not increase radioresistance and made JJ012 more radiosensitive.
- **Variance:** vs unirradiated cells; X-ray irradiation as the comparator for carbon ions; normoxic culture as comparator for hypoxia; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** In-vitro only, and the hypoxia result cuts against the standard oxygen-enhancement-ratio argument for why carbon ions should be preferred in this histology.

Vares/Saintigny (2020) — Radiother Oncol

Reference:PMID 32717360 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Carbon ions with miR-34 mimic and rapamycin (cancer stem cell fraction)
- **Indication / model:** CH-2879 high-grade chondrosarcoma cells, cancer-stem-cell and tumour-initiating assays
- **Design:** A radioresistant stem-like fraction survives irradiation; mTOR inhibition raises FOXO3 and miR-34, which represses KLF4 and strips the stemness programme that lets those cells re-initiate the tumour.
- **n:** single cell line; replicate counts not stated in the abstract
- **Effect size:** moderate — High-grade chondrosarcoma cultures contained a radioresistant cancer-stem-cell population that carbon ions alone did not clear, and the triple combination controlled it along with the stemness phenotype and tumour-initiating capacity.
- **Variance:** vs carbon-ion irradiation alone; untreated cells; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Single conventional grade III line and no animal arm; the miR-34 mimic has no clinical route in sarcoma.

Richert/Dutour (2020) — J Bone Oncol

Reference:PMID 31956474 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Immune landscape of conventional versus dedifferentiated chondrosarcoma
- **Indication / model:** human tumour cohort, IHC (CD3, CD8, CD68, CD163, PD-L1, CSF1R) plus RT-qPCR for immune checkpoints
- **Design:** The dominant immune population is macrophage, not lymphocyte, and the checkpoints that are transcribed (CSF1R, SIRPA alongside B7H3, TIM3, LAG3) are largely myeloid ones, so PD-L1 positivity marks an inflamed dedifferentiated component without being the switch that controls it.
- **n:** n=76 (27 conventional, 49 dedifferentiated)
- **Effect size:** moderate — TAMs were the main immune population in both subtypes; a high CD68+/CD8+ ratio was an independent poor prognostic factor ($p<0.01$) and high CD68+ tracked with metastasis at diagnosis ($p<0.05$). PD-L1 protein was confined to dedifferentiated tumours (42.6%), and CSF1R was expressed by TAMs in 89.7% of dedifferentiated versus 62.9% of conventional cases.
- **Variance:** vs conventional chondrosarcoma as the comparator for the dedifferentiated group; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Descriptive human-tissue study with no therapeutic arm, so the CSF1R nomination is a hypothesis. PubMed indexes the author names in scrambled form; the 2020 corrigendum (PMID 32274326) gives them as Richert I ... Dutour A, which is what is recorded here.

—/— (2020) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Ivosidenib (AG-120)
- **Indication / model:** IDH1-mutant chondrosarcoma, conventional grade 2 or 3 (2L+ (progression within 4 months required)); IDH1 mutation, CLIA-confirmed; central pathology review
- **Design:** Single-arm phase 2
- **n:** 6

- **Effect size:** — — Progression-free survival
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2020) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** LY3410738 (covalent mutant IDH1/IDH2 inhibitor)
- **Indication / model:** Advanced solid tumours with IDH1 or IDH2 mutations (cholangiocarcinoma, chondrosarcoma, glioma named) (2L+); IDH1 R132 by tissue or ctDNA (ctDNA accepted for chondrosarcoma)
- **Design:** Phase 1 dose escalation and expansion, biomarker-selected basket
- **n:** 200
- **Effect size:** — — Recommended phase 2 dose
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2020) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Olaparib plus ceralasertib (AZD6738)
- **Indication / model:** IDH1 or IDH2 mutant solid tumours (non-cholangiocarcinoma, non-CNS) (2L+ (progression despite standard therapy required)); IDH1 or IDH2 mutation with neomorphic activity, CLIA-confirmed
- **Design:** Simon two-stage single-arm phase 2, biomarker-selected basket
- **n:** 24
- **Effect size:** — — Objective response rate
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Marabelle A/Bang YJ (2020) — Lancet Oncol

Reference:PMID 32919526 · **Type:** trial · **Inclusion:** included

- **Intervention:** Pembrolizumab 200 mg IV every 3 weeks
- **Indication / model:** Advanced solid tumours, tissue TMB-high (KEYNOTE-158 biomarker analysis) (2L+ (progression on or intolerance to at least one prior line)); tissue TMB >=10 mutations per megabase (FoundationOne CDx)
- **Design:** Prospective biomarker analysis within a multi-cohort single-arm phase 2; 790 of 1050 patients evaluable for tissue TMB, 102 tTMB-high
- **n:** 790
- **Effect size:** 29 — Objective response rate by independent central review: 29% (95% CI 21-39) in tTMB-high versus 6% (95% CI 5-8) in non-tTMB-high
- **Variance:** 95% CI 21–39
- **Case fit:** partial

- **Notes:** (toxicity table not extracted)

—/— (2020) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Sarcoma-specific CAR-T cells including anti-CD276 (B7-H3), with low-dose chemotherapy and irradiated tumour-cell vaccine
- **Indication / model:** Stage III-IV or recurrent sarcoma with confirmed CAR-T target antigen expression (any (at least 2 weeks from last chemotherapy or radiotherapy)); surface antigen expression by IHC or flow: GD2, PSMA, HER2, CD276 (B7-H3) or other
- **Design:** Open-label phase 1/2, multiple CAR-T products plus low-dose chemotherapy and maintenance vaccine
- **n:** 20
- **Effect size:** — — Safety of CAR-T infusion
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Song/Jiang (2019) — Clin Orthop Relat Res

Reference:PMID 30762691 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Resection of the primary in metastatic chondrosarcoma (SEER)
- **Indication / model:** Metastatic chondrosarcoma at diagnosis, all subtypes (1L); 200 SEER patients 1988-2014; multivariable Cox with interaction tests by subtype and grade.
- **Design:** Retrospective population-based cohort (SEER)
- **n:** 200
- **Effect size:** 0.481 HR — OS hazard ratio, resection vs none
- **Variance:** 95% CI 0.34–0.68; p=<0.001
- **Toxicities:** SEER carries no AE or complication data.
- **Case fit:** partial
- **Notes:** The counterweight to the Japanese registry: in the subgroup that is actually hers - dedifferentiated, high-grade - resection of the primary showed no survival association. The oncologic case for resecting her acetabular primary is unproven; the structural case for stabilising the fracture is a separate decision this literature does not answer.

Wang/Ye (2019) — J Cancer

Reference:PMID 31258751 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Prognostic factors in metastatic chondrosarcoma (SEER)
- **Indication / model:** Primary chondrosarcoma of bone with metastatic disease at diagnosis (1L); 264 SEER patients 2000-2013, all subtypes and grades pooled (majority conventional).
- **Design:** Retrospective population-based cohort (SEER)
- **n:** 264
- **Effect size:** 14.0 months — Median OS, metastatic at diagnosis
- **Variance:** ±2.5 (SE)
- **Toxicities:** SEER carries no AE data.

- **Case fit:** partial
- **Notes:** Pooled-subtype metastatic figures: the 14-month median OS is dominated by conventional and lower-grade disease and overstates her expectation - dedifferentiated-specific metastatic medians run 5-7 months (Grimer, Bui, Masunaga). Kept as the denominator-labelled contrast the honesty rule requires.

Nakagawa/Kitabayashi (2019) — Oncogene

Reference:PMID 31406254 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** DS-1001b, selective mutant-IDH1 inhibitor (in-vivo evidence)
- **Indication / model:** IDH1-mutant chondrosarcoma cell lines (conventional L835, JJ012) in vitro and in mouse xenografts
- **Design:** Mutant IDH1 inhibition demethylates H3K9me3 and restores SOX9 and CDKN1C, which pushes conventional cells back toward chondrocyte differentiation and arrests the cycle in the more dedifferentiated line.
- **n:** not stated per arm in the abstract
- **Effect size:** moderate — DS-1001b lowered 2-HG and impaired proliferation of IDH1-mutant chondrosarcoma cells in vitro and in vivo, with the transcriptional response splitting by cell type: differentiation in L835, cell-cycle arrest in JJ012.
- **Variance:** vs vehicle; IDH1 wild-type comparators; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** The strongest in-vivo case for a mutant-IDH1 inhibitor in this histology, but the effect is growth impairment rather than regression, and JJ012 is treated here as the dedifferentiated surrogate on thin grounds.

Venneker/Bovée (2019) — Cancers (Basel)

Reference:PMID 31810230 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Talazoparib alone and with temozolomide or radiation
- **Indication / model:** chondrosarcoma cell lines with and without endogenous IDH1/IDH2 mutation, including lines carried long-term on a D-2-HG-normalising inhibitor
- **Design:** The IDH-mutant 'BRCAness' hypothesis holds that D-2-HG suppresses homologous recombination and creates PARP-inhibitor synthetic lethality, as it does in glioma and leukaemia models.
- **n:** cell-line panel; counts not stated in the abstract
- **Effect size:** moderate — Talazoparib sensitivity varied widely and did not sort by IDH mutation status, and long-term normalisation of D-2-HG did not create synthetic lethality. A subset of lines was impaired in PARylation capacity, homologous recombination and MGMT expression, and talazoparib synergised with temozolomide or radiation independently of mutant-IDH1 inhibition.
- **Variance:** vs IDH wild-type lines; single-agent temozolomide or radiation; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** In-vitro only. It supports PARP-inhibitor combinations in chondrosarcoma while removing IDH status as the selection biomarker for them, which matters for how the olaparib-plus-ceralasertib trial would be entered.

Césaire/Chevalier (2019) — J Bone Oncol

Reference:PMID 31312595 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Olaparib with X-ray, proton or carbon-ion irradiation
- **Indication / model:** CH2879 grade III chondrosarcoma cell line
- **Design:** PARP inhibition blocks single-strand break repair, so the clustered damage from particles is converted into lethal double-strand breaks the cell cannot resolve.
- **n:** single cell line; clonogenic and proliferation assays
- **Effect size:** moderate — Olaparib gave sensitiser enhancement ratios of roughly 1.3 with X-rays, 1.8 with protons and 1.5 with carbon ions on clonogenic assay. CH2879 carried mutations in RAD50, SMARCA2 and NBN, which the authors link to its baseline radiosensitivity.
- **Variance:** vs irradiation without olaparib; unirradiated cells; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Single cell line, in-vitro only, and no dedifferentiated model. The largest gain was with protons rather than carbon ions, which is not the ordering the clinical modality argument assumes.

Majzner/Mackall (2019) — Clin Cancer Res

Reference:PMID 30655315 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** B7-H3 (CD276) CAR T cells
- **Indication / model:** xenograft models of osteosarcoma, Ewing sarcoma and medulloblastoma; antigen-density titration in vitro
- **Design:** CAR engagement is a threshold phenomenon rather than a binary one: below a certain surface antigen count per cell the synapse does not trigger full effector function, which is what creates a therapeutic window against normal tissue that expresses B7-H3 at low level.
- **n:** multiple xenograft models; per-arm counts not stated in the abstract
- **Effect size:** strong — B7-H3 CAR T cells regressed established solid tumours including osteosarcoma and Ewing sarcoma xenografts. Efficacy depended on high surface antigen density and was greatly diminished against low-expressing targets.
- **Variance:** vs untransduced or control CAR T cells; low- versus high-antigen target cells; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** No chondrosarcoma model anywhere in the paper; the density-dependence result is the part that transfers, and it cuts against a tumour whose measured B7-H3 H-scores are the lowest in sarcoma.

Ouyang/Liu (2019) — Cell Commun Signal

Reference:PMID 30808351 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Palbociclib (CDK4/6 inhibition) in chondrosarcoma
- **Indication / model:** human chondrosarcoma cell lines with CDK4 siRNA and palbociclib; intraosseous chondrosarcoma mouse model read out by bioluminescence
- **Design:** CDK4 phosphorylates Rb to release the G1 restriction point; where CDKN2A is deleted, as it is in a large share of dedifferentiated chondrosarcoma, that brake is already gone and inhibiting CDK4 restores it pharmacologically.
- **n:** not stated per arm in the abstract
- **Effect size:** moderate — CDK4 was expressed in human chondrosarcoma samples and correlated with metastasis and worse prognosis; siRNA or palbociclib cut proliferation with reduced Rb phosphorylation, arrested cells in G1 and reduced migration and invasion. Palbociclib reduced tumour burden within bone in vivo.

- **Variance:** vs scrambled siRNA and vehicle-treated mice; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** A 2023 correction replaced a wrongly uploaded Fig. 5C image with conclusions unchanged (PMID 36803557). CDKN2A status of the lines is not reported, so the biomarker link to this patient's untested CDKN2A remains inferred.

—/— (2019) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Abemaciclib
- **Indication / model:** Advanced bone and soft tissue sarcoma with CDK-pathway alteration; dedifferentiated chondrosarcoma named (2L+ for dedifferentiated chondrosarcoma (at least one prior anthracycline required)); CCND1/2/3, CDK4 or CDK6 amplification or gain, or CDKN2A/B loss; Rb-positive by central IHC
- **Design:** Two-step single-arm phase 2
- **n:** 44
- **Effect size:** — — Progression-free survival
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Lex/Grimer (2018) — Clin Sarcoma Res

Reference:PMID 30559960 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Surgery for pelvic dedifferentiated chondrosarcoma
- **Indication / model:** Dedifferentiated chondrosarcoma arising in the pelvis (1L); 31 pelvic DDCS, 1995-2016, single UK centre; 41.9% presented locally or systemically advanced; wide margin defined as >4 mm.
- **Design:** Retrospective single-centre cohort
- **n:** 31
- **Effect size:** 50% vs 15.4% % — 12-month OS, surgical vs palliative intent
- **Toxicities:** No structured complication reporting in the abstract.
- **Case fit:** strong
- **Notes:** The closest anatomical match in the dossier - dedifferentiated disease in the pelvis. The margin price of oncologic surgery there was hindquarter amputation, and the comparison with palliative care is confounded by fitness and disease extent, which is precisely her situation. Oncologic clearance and structural stabilisation are different operations; this paper speaks to the first.

D'Angelo/Streicher (2018) — Lancet Oncol

Reference:PMID 29370992 · **Type:** clinical · **Inclusion:** excluded — No chondrosarcoma-specific outcome data: Alliance A091401 enrolled a predominantly soft-tissue sarcoma population and reports no chondrosarcoma subset, so it adds nothing beyond SARC028 and IMMUNOSARC for her histology.

- **Intervention:** Nivolumab with or without ipilimumab (Alliance A091401)
- **Notes:** Excluded: No chondrosarcoma-specific outcome data: Alliance A091401 enrolled a predominantly soft-tissue sarcoma population and reports no chondrosarcoma subset, so it adds nothing beyond SARC028

and IMMUNOSARC for her histology.

Peterse/Bovée (2018) — Br J Cancer

Reference:PMID 29576625 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CB-839 / metformin / phenformin / chloroquine (glutaminolysis)
- **Indication / model:** 10 chondrosarcoma cell lines with and without endogenous IDH1/2 mutation; glutaminase IHC on 120 cartilage tumours
- **Design:** Glutaminase feeds the TCA cycle when isocitrate handling is disturbed; the prediction was that IDH-mutant cells would be the sensitive ones.
- **n:** 10 cell lines; 120 tumour samples for IHC
- **Effect size:** moderate — Glutaminase expression rose with tumour grade (p=0.001), highest in high-grade chondrosarcoma, and all four agents had activity — but sensitivity did not track IDH1/2 mutation status. Metformin and phenformin lowered mTOR activity, and metformin's effect on LC3B-II was reversed by chloroquine.
- **Variance:** vs IDH wild-type chondrosarcoma lines; untreated cells; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** In-vitro only, no in-vivo arm. The grade association is the transferable part; the IDH-stratification hypothesis that motivated the glutaminase trials is not supported here.

Tawbi/Patel (2017) — Lancet Oncol

Reference:PMID 28988646 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pembrolizumab (SARC028, bone-sarcoma cohort)
- **Indication / model:** Advanced bone sarcoma (osteosarcoma 22, Ewing 13, chondrosarcoma 5 evaluable) (2L+); Up to 3 prior lines; biopsy-accessible lesion required; bone cohort 42 treated, 40 evaluable, including 5 chondrosarcoma without subtype restriction.
- **Design:** Two-cohort single-arm phase 2 (SARC028, NCT02301039)
- **n:** 84
- **Effect size:** 20 % — ORR, chondrosarcoma subgroup
- **Toxicities:** anaemia G>=3 (6/42, 14%); lymphocyte count decreased G>=3 (5/42, 12%); activated partial thromboplastin time prolonged G>=3 (4/42, 10%); platelet count decreased G>=3 (3/42, 7%); treatment-emergent serious AE (5/42, 12%); pneumonitis (immune-related SAE) (2/84, 2.4%)
- **Case fit:** weak
- **Notes:** Origin of the claim that checkpoint activity in chondrosarcoma clusters in the dedifferentiated variant; the primary paper does not actually subtype the single responder, and later sources attribute dedifferentiated histology secondhand. One responder among five patients is the entire prospective pembrolizumab evidence base in this disease.

Voissiere/Miot-Noirault (2017) — PLoS One

Reference:PMID 28704566 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Doxorubicin and hypoxia-activated prodrug TH-302 in a hypoxic spheroid
- **Indication / model:** HEMC-SS human chondrosarcoma multicellular tumour spheroids, day 7 (non-hypoxic) vs

day 14/20 (hypoxic core)
- **Design:** Spheroid growth reproduces two features of the tumour at once: a glycosaminoglycan- and collagen-II-rich matrix that blocks drug penetration, and a pimonidazole-positive hypoxic core with VEGF secretion.
- **n:** spheroids assayed at three time points; replicate counts not stated in the abstract
- **Effect size:** moderate — Doxorubicin resistance rose with spheroid size (multicellular resistance index 5.0 at day 7 vs 18.3 at day 14), and doxorubicin autofluorescence confirmed it was not reaching the deep layers. The hypoxia-activated prodrug TH-302 moved the other way, working better on the larger hypoxic spheroids.
- **Variance:** vs matched 2D monolayer culture of the same line, and day-7 spheroids as the non-hypoxic comparator; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Single cell line, extraskeletal myxoid rather than conventional or dedifferentiated central chondrosarcoma, and no in-vivo arm.

Simard/Dutour (2017) — Oncoimmunology

Reference:PMID 28344871 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Selective T-cell versus CD163+ macrophage depletion (syngeneic model)
- **Indication / model:** Swarm rat chondrosarcoma (SRC) syngeneic model plus IHC on human chondrosarcoma samples
- **Design:** Causal rather than correlative: removing the two populations in opposite directions separates which one restrains the tumour and which one supports it.
- **n:** not stated per arm in the abstract
- **Effect size:** moderate — Depleting T lymphocytes accelerated tumour growth, while depleting CD163+ macrophages slowed it. Splenocytes from tumour-bearing rats showed 27% specific cytotoxicity against chondrosarcoma cells, falling to 11% after CD3 depletion, and in both human and rat tumours the immune infiltrate sat in the peritumoral area.
- **Variance:** vs undepleted tumour-bearing rats; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Rat chondrosarcoma rather than a dedifferentiated human model; the immunocompetent design is what makes it worth keeping despite that.

Kostine/Bovée (2016) — Mod Pathol

Reference:PMID 27312065 · **Type:** clinical · **Inclusion:** included

- **Intervention:** PD-L1 expression in chondrosarcoma (biomarker context)
- **Indication / model:** Chondrosarcoma, all subtypes (tissue microarray + whole-section validation) (any); TMA: 119 conventional, 19 mesenchymal, 20 clear-cell, 22 dedifferentiated; independent whole-section dedifferentiated cohort n=21.
- **Design:** Immunohistochemical biomarker study
- **n:** 181
- **Effect size:** 41% (9/22) TMA; 52% (11/21) whole-section % — PD-L1 positivity, dedifferentiated subtype
- **Toxicities:** Not applicable (no therapeutic exposure).
- **Case fit:** strong

- **Notes:** Source of the 'PD-L1 in roughly half of dedifferentiated tumours' figure; PD-L1 sits in the dedifferentiated component, so her PD-L1 assay should be run on that component. Prevalence evidence only - it does not by itself predict response.

Outani/Yoshikawa (2016) — Int J Clin Oncol

Reference:PMID 26150259 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Carbon-ion radiotherapy vs surgery, pelvic chondrosarcoma
- **Indication / model:** Pelvic chondrosarcoma treated with curative intent (1L); 31 patients, 1983-2014, non-metastatic; median age 43; median follow-up 66 months.
- **Design:** Retrospective comparative cohort (surgery vs CIRT)
- **n:** 31
- **Effect size:** 72 % — 5-year OS, whole cohort
- **Toxicities:** No per-term complication table in the source abstract; functional outcome favoured CIRT.
- **Case fit:** partial
- **Notes:** Already carried by the trial screen; re-logged here as clinical evidence with the caveat that the cohort was non-metastatic and curative-intent, so survival figures do not transfer - what transfers is the local-control-with-preserved-function comparison for a periacetabular lesion. Only 7 patients received CIRT.

Hamdi/Saintigny (2016) — Int J Radiat Oncol Biol Phys

Reference:PMID 27084635 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Carbon ions versus X-rays in a normal-cartilage 3D model (normal-tissue side)
- **Indication / model:** primary human articular chondrocytes from a healthy donor in a collagen scaffold under physioxia (3D), with 2D culture as reference
- **Design:** Entrance-channel carbon ions cross healthy cartilage before the Bragg peak; senescence rather than death is the relevant injury endpoint in a tissue that does not turn over.
- **n:** one healthy donor; survival, cell-death and senescence assays
- **Effect size:** weak — In 2D the carbon-ion RBE was 2.6 with a two-fold increase in radiation-induced senescence, but in the 3D physioxic cartilage model carbon ions and X-rays induced senescence at comparable rates. Measuring RBE in monolayer may therefore misstate the normal-tissue cost of a carbon-ion plan.
- **Variance:** vs X-ray irradiation at the same dose; 2D culture as the reference format; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Healthy donor cartilage, not tumour, and a single donor. Speaks to the joint-preservation side of an acetabular particle-therapy decision rather than to tumour control.

Ogitani/Agatsuma (2016) — Cancer Sci

Reference:PMID 27166974 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** DXd-payload bystander killing (DS-8201a, mechanism read-across)
- **Indication / model:** co-culture of antigen-positive and antigen-negative cells; mouse xenografts with mixed antigen-positive and antigen-negative tumours
- **Design:** The released DXd payload is membrane-permeable, so it diffuses out of the antigen-positive cell it was delivered into and kills adjacent antigen-negative neighbours; the non-cleavable T-DM1 payload does not do

this.
- **n:** co-culture and mixed-xenograft experiments; per-arm counts not stated in the abstract
- **Effect size:** moderate — DS-8201a killed antigen-negative cells in co-culture and in mixed xenografts, while T-DM1 did not. This is the mechanism by which a DXd ADC can act on a heterogeneous or low-expressing tumour rather than only on strongly positive cells.
- **Variance:** vs T-DM1, whose non-cleavable payload does not diffuse; antigen-negative tumours alone; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** HER2 breast/gastric models, a different antigen and a different tumour entirely; it is offered as the payload-class rationale for why a B7-H3 DXd ADC might not need uniform high antigen, not as evidence that it works in chondrosarcoma.

—/— (2016) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Carbon-ion radiotherapy
- **Indication / model:** Unresectable or R2-resected radioresistant tumours, including chondrosarcoma WHO grade >=2 outside the skull base (local therapy, any systemic line); none required
- **Design:** Transnational randomised trial of carbon-ion therapy versus conventional radiotherapy including protons (ETOILE)
- **n:** 250
- **Effect size:** — — Progression-free survival
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2016) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Marketed targeted agents assigned by genomic match
- **Indication / model:** Advanced solid tumours, lymphoma or myeloma with an actionable genomic alteration (2L+ (no longer benefiting from, or no available, standard treatment)); genomic alteration matched to a marketed targeted agent, from any CLIA panel
- **Design:** Non-randomised multi-cohort phase 2 basket registry (TAPUR)
- **n:** 4200
- **Effect size:** — — Objective response or stable disease at 16 weeks
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Suijker/Bovée (2015) — Oncotarget

Reference:PMID 25895133 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Mutant-IDH1 inhibition (AGI-5198) in chondrosarcoma lines

- **Indication / model:** chondrosarcoma cell lines with endogenous IDH1 (n=3) or IDH2 (n=2) mutation, versus IDH wild-type lines
- **Design:** If mutant IDH were still driving the malignant phenotype, removing D-2-HG should reverse the epigenetic block it causes. It did not, which places the mutation upstream of an event the cell has already banked.
- **n:** 5 IDH-mutant lines; treatment to 72 h and prolonged culture up to 20 passages
- **Effect size:** negative — AGI-5198 dropped D-2-HG by more than 90% but only one of three mutant IDH1 lines showed even a moderate viability decrease at 72 h, and proliferation and migration were unchanged out to 20 passages. Global gene expression, CpG island methylation and H3K4/K9/K27 trimethylation all stayed put.
- **Variance:** vs IDH1/2 wild-type chondrosarcoma cell lines and vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** In-vitro only, and the lines are conventional central chondrosarcoma rather than dedifferentiated, so this is the founder-event argument rather than a direct dedifferentiated-subtype result.

Li/Trent (2015) — PLoS One

Reference:PMID 26368816 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Mutant-IDH1 inhibition (AGI-5198), opposing result
- **Indication / model:** JJ012 (IDH1 R132G) and HT1080 (IDH1 R132C) cell lines; C28 IDH wild-type chondrocytes
- **Design:** Same premise as Suijker: block the mutant enzyme, lose D-2-HG, lose the transformed phenotype.
- **n:** 3 cell lines; replicate counts not stated in the abstract
- **Effect size:** moderate — AGI-5198 lowered D-2-HG dose-dependently and cut colony formation and migration, interrupted cycling and induced apoptosis. D-2-HG was also detectable in plasma and urine of an IDH-mutant chondrosarcoma patient and fell after resection.
- **Variance:** vs IDH1 wild-type C28 chondrocytes and vehicle; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** HT1080 is a fibrosarcoma line, not chondrosarcoma, which weakens half the panel; in-vitro only. Directly conflicts with Suijker 2015, and the ivosidenib clinical result in dedifferentiated disease sits closer to Suijker.

Italiano/Blay (2013) — Ann Oncol

Reference:PMID 24099780 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Anthracycline-based chemotherapy (advanced chondrosarcoma)
- **Indication / model:** Advanced chondrosarcoma, all subtypes (88% metastatic) (any); 180 patients from 15 institutions, 1988-2011; 63% conventional histology, dedifferentiated subset n=44 by the subtype response table. Median age 52.
- **Design:** Multicentre retrospective cohort (French Sarcoma Group)
- **n:** 180
- **Effect size:** 20.5 % — ORR by RECIST, dedifferentiated subset
- **Variance:** p=0.04
- **Toxicities:** No per-term toxicity reporting; retrospective chart review of heterogeneous regimens.
- **Case fit:** strong
- **Notes:** The anchor row for the chemotherapy fork: the dedifferentiated ORR of 20.5% is why cytotoxics are on the table at all, and the PS>=2 finding cuts directly at her assumed ECOG 2. mPFS/mOS are pooled across

subtypes, not dedifferentiated-specific. No AE data in the source (retrospective registry); abstract and PubMed record checked, no trial registration.

Monderer/Blanchard (2013) — Lab Invest

Reference:PMID 23958880 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Doxorubicin / mafosfamide (matrix-mediated chemoresistance mechanism)
- **Indication / model:** two new patient-derived chondrosarcoma cell lines (one dedifferentiated, one grade III conventional central) in monolayer vs 3D chondrogenic pellet; nude-mouse transplantation of biopsies and one cell line
- **Design:** Cartilaginous matrix laid down in 3D pellet culture slows diffusion of drug to the cell, so nuclear drug accumulation falls and the same concentration that kills in monolayer no longer does.
- **n:** 10 biopsies collected; 1 transplanted biopsy and 1 injected cell line engrafted; 2 stable cell lines derived
- **Effect size:** moderate — The same cells were sensitive to conventional agents in monolayer and resistant to low-dose mafosfamide or doxorubicin once they had rebuilt a hyaline cartilage matrix in 3D pellets, with altered nuclear drug accumulation. Both lines carried IDH1, IDH2 and TP53 mutations with CDKN2A deletion and/or MDM2 amplification, matching the genotype spectrum of the patient's likely tumour.
- **Variance:** vs matched monolayer (2D) culture of the same lines; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Culture-format comparison rather than an in-vivo treatment arm; the engraftment yield of 1 in 10 biopsies is itself a statement about how hard this tumour is to model.

Nicholls/Ferguson (2013) — J Orthop Res

Reference:PMID 23606416 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Radiotherapy and periosteal stripping before femoral fracture (bone-healing mechanism)
- **Indication / model:** female Wistar retired-breeder rat femoral fracture with fixation, after radiotherapy and/or surgical periosteal stripping
- **Design:** Irradiated bone loses the chondroid and osteoid production that callus formation depends on, so the fracture defaults to atrophic non-union regardless of the mechanical fixation.
- **n:** 4 treatment groups (control, radiotherapy, surgery, combined) across 3 endpoints (21, 28, 35 days post-fracture)
- **Effect size:** strong — By 35 days controls had united, periosteum-stripped animals formed hypertrophic non-unions, and irradiated animals formed atrophic non-unions with histomorphometry showing absent chondroid and osteoid. Periosteal stripping added little; radiation alone was sufficient.
- **Variance:** vs untreated fracture; surgery-alone and radiotherapy-alone arms; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Normal rat bone with conventional photons and no tumour present, so it speaks to healing of the fractured acetabulum after radiotherapy rather than to tumour biology or drug delivery at that site.

van Oosterwijk/Bovée (2012) — Ann Oncol

Reference:PMID 22112972 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Doxorubicin / cisplatin with BH3 mimetic ABT-737 (apoptotic-block mechanism)
- **Indication / model:** chondrosarcoma cell lines SW1353, CH2879, JJ012, OUMS27 plus two primary cultures; 3D pellet models
- **Design:** Resistance sits downstream of drug delivery: BCL-2 family members hold the mitochondrial apoptotic gate shut, and a BH3 mimetic reopens it so that DNA damage is converted into cell death.
- **n:** 4 cell lines + 2 primary cultures
- **Effect size:** strong — Doxorubicin reached the nuclei of cells inside 3D pellets despite the cartilaginous matrix, and MDR pump activity was inconsistent across cultures, which undercuts both of the usual explanations. Adding ABT-737 to doxorubicin or cisplatin abolished viability with cytochrome C release and apoptosis.
- **Variance:** vs single-agent doxorubicin or cisplatin, and untreated cells; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** No in-vivo arm, and the panel is dominated by conventional grade II-III lines rather than dedifferentiated ones. Directly contradicts the matrix-diffusion result in Monderer 2013 on the point of nuclear drug accumulation.

Puri/Basappa (2009) — J Cancer Res Ther

Reference:PMID 19293483 · **Type:** clinical · **Inclusion:** excluded — Wrong population for the fracture question: 45 non-metastatic, mostly conventional chondrosarcomas from a single centre; the pathologic-fracture question is answered in dedifferentiated disease directly by Sambri 2021 and Grimer 2007.

- **Intervention:** Chondrosarcoma of bone: size, pathologic fracture and prior intervention
- **Notes:** Excluded: Wrong population for the fracture question: 45 non-metastatic, mostly conventional chondrosarcomas from a single centre; the pathologic-fracture question is answered in dedifferentiated disease directly by Sambri 2021 and Grimer 2007.

Kim/Lee (2009) — Mol Cancer

Reference:PMID 19445670 · **Type:** preclinical · **Inclusion:** excluded — In-vitro only, in two grade II conventional lines (JJ012, SW1353); the same anti-apoptotic mechanism is covered with a druggable BH3 mimetic and a matrix-relevant 3D model in van Oosterwijk 2012.

- **Intervention:** Doxorubicin with BCL-2/BCL-xL/XIAP or P-glycoprotein knockdown
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: In-vitro only, in two grade II conventional lines (JJ012, SW1353); the same anti-apoptotic mechanism is covered with a druggable BH3 mimetic and a matrix-relevant 3D model in van Oosterwijk 2012.

Schrage/Bovée (2009) — J Cell Mol Med

Reference:PMID 18624751 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** pRb-pathway restoration: p16 overexpression and CDK4 knockdown
- **Indication / model:** chondrosarcoma cell lines OUMS27, SW1353, CH2879 with CDKN2A/p16 overexpression vectors and shRNA against CDK4; qPCR and IHC on 34 fresh-frozen and 90 FFPE enchondroma and chondrosarcoma samples
- **Design:** Reintroducing p16 or removing CDK4 rebuilds the same G1 brake that CDKN2A deletion removes during progression, which tests whether the pathway is still load-bearing rather than merely altered.
- **n:** 3 cell lines; 124 human tumour samples; IHC subset of 29 high-grade chondrosarcomas

- **Effect size:** moderate — Both p16 overexpression and CDK4 knockdown inhibited chondrosarcoma cell growth in vitro, and pRb-pathway alterations tracked with progression from enchondroma to high-grade chondrosarcoma in the tissue series.
- **Variance:** vs empty vector and non-targeting shRNA; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** In-vitro genetic proof rather than drug testing, and predates the availability of CDK4/6 inhibitors; establishes that the pathway is functional, not that a given inhibitor works.

Grimer/Ferrari (2007) — Eur J Cancer

Reference:PMID 17720491 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Surgery-centred multimodal management (EMSOS series)
- **Indication / model:** Dedifferentiated chondrosarcoma, nine European centres (1L); 337 patients; 71 (21%) metastatic at diagnosis; 29% of long-bone tumours presented with pathological fracture.
- **Design:** Multicentre retrospective cohort (EMSOS)
- **n:** 337
- **Effect size:** 5 months — Median OS, metastatic at diagnosis
- **Toxicities:** No per-term AE or surgical-complication table.
- **Case fit:** strong
- **Notes:** Largest dedifferentiated cohort ever assembled and the honest baseline for her presentation: metastatic at diagnosis carried a median survival of 5 months. Both of her adverse anatomy features - pathological fracture and pelvic site - were independent poor prognostic factors in the non-metastatic analysis. Corresponding email is the 2007 published address of the first author.

Morioka/Hornicek (2003) — Clin Cancer Res

Reference:PMID 12631627 · **Type:** preclinical · **Inclusion:** excluded — The antiangiogenic partner (plasminogen-related protein B) never reached clinical development, so the combination is not an option that could be acted on; the low-vascularity argument it speaks to is carried by newer models with usable agents.

- **Intervention:** Trabectedin (ET-743) with recombinant plasminogen-related protein B
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: The antiangiogenic partner (plasminogen-related protein B) never reached clinical development, so the combination is not an option that could be acted on; the low-vascularity argument it speaks to is carried by newer models with usable agents.

Mitchell/Tillman (2000) — J Bone Joint Surg Br

Reference:PMID 10697315 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Planned resection plus chemotherapy (Birmingham series)
- **Indication / model:** Dedifferentiated chondrosarcoma, single UK centre (1L); 22 consecutive patients treated with planned resection and chemotherapy at the Royal Orthopaedic Hospital.
- **Design:** Retrospective single-centre series
- **n:** 22
- **Effect size:** <9 months — Median OS

- **Toxicities:** No per-term AE reporting.
- **Case fit:** partial
- **Notes:** Early series often cited for the chemo-plus-wide-margins signal; likely overlaps the later Grimer EMSOS pooled cohort (same centre). No AE data in source; abstract and record checked.